

Roll No.

TEC-503

B. TECH. (ECE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
MICROCONTROLLER AND EMBEDDED SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Draw and explain the block diagram of 8051 microcontroller. (CO1)

OR

- (b) Explain the specifications needed for selecting any microcontroller based embedded systems. Mention example of embedded systems.

(CO1)

2. (a) Explain the different types for microcontroller. (CO1)

OR

- (b) Explain the difference between Von-Neumann and Harvard architecture.

(CO1)

3. (a) Draw the PSW of 8051 microcontroller. Write a program to add – 128 and – 2 and mention the flags and accumulator after the execution of the program. (CO2)

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OR

- (b) Explain the memory organization of 8051 microcontroller. (CO2)
4. (a) Write a program by 8051 to verify the result of signed arithmetic operation in RAM location 27H that if the stored result is positive or negative. If it is negative send a high value of P1.7, otherwise send a low value. (CO2)

OR

- (b) Analyze the entire code and answer step by step : (CO2)

```
MOV A,#42H
```

```
ADD A,#13H
```

```
DA A
```

```
ADD A,#17H
```

```
DA A
```

```
ADDC A,#34H
```

```
DA A
```

```
ADDC A,#11H
```

```
DA A
```

```
END
```

5. (a) Write a program to find a number when XORed with A register, result in number 3FH in A. Prove it also by example. (CO2)

(3)

OR

(b) Analyze the code and answer step by step :

(CO2)

MOV D0H,#00H

MOV A,#3AH

MOV 45H,#13H

SUBB A,45H

SUBB A,#80H

SUBB A,#22H

SUBB A,#FFH

END